Supplemental Materials

HiBER: Hierarchical Bounded Strength Envelopes for Rockfill and Mine Waste Rock

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This file contains Figs. S1–S5, Tables S1–S6, and a worked example of the update, cited in the paper. Equation and table numbers without the prefix S refer to the paper.

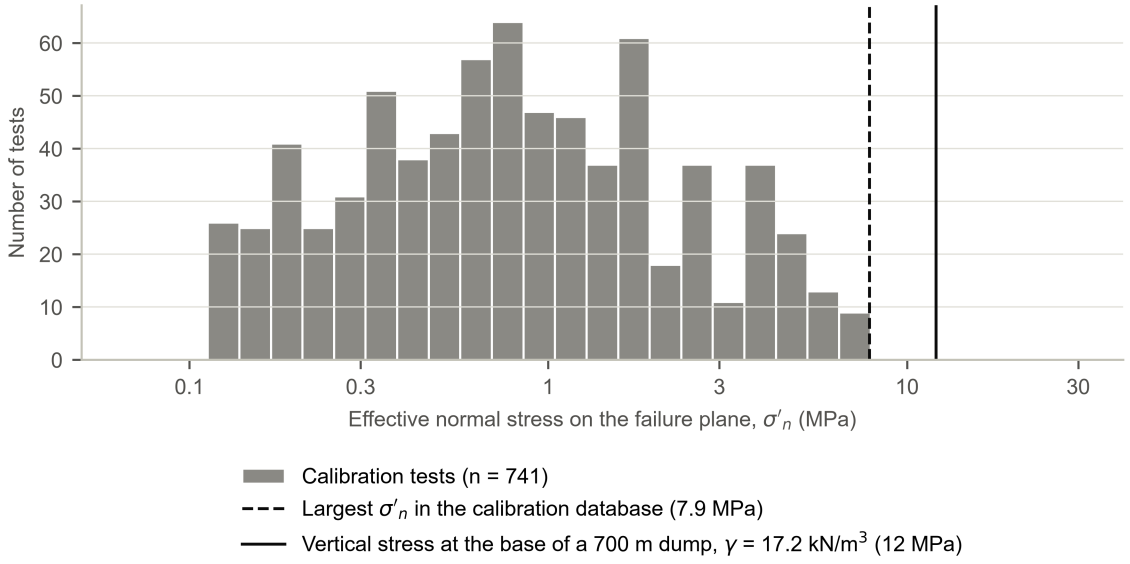


Fig. S1. Distribution of effective normal stress on the failure plane in the calibration database (741 tests).

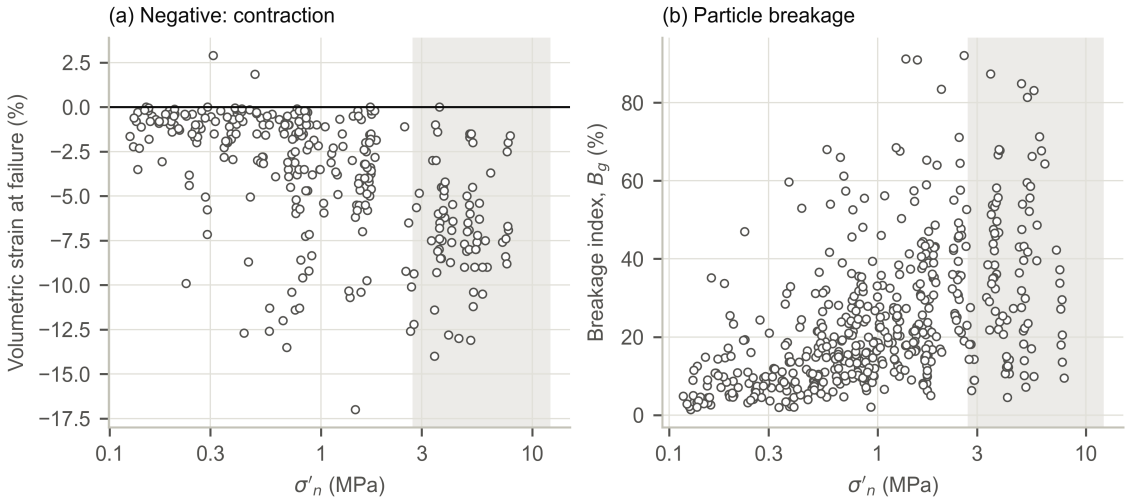


Fig. S2. Net volumetric strain at the maximum deviator (negative: contraction) and breakage index of Marsal (1973) versus σ'_n in the calibration database. Reported breakage indices above 100%, impossible by definition, were excluded (4 tests). Above 1.5 MPa the residuals of HiBER increased with the volumetric compression (Spearman ρ = 0.38, n = 122) and, above 2.75 MPa, with the breakage index (ρ = 0.24, n = 91).

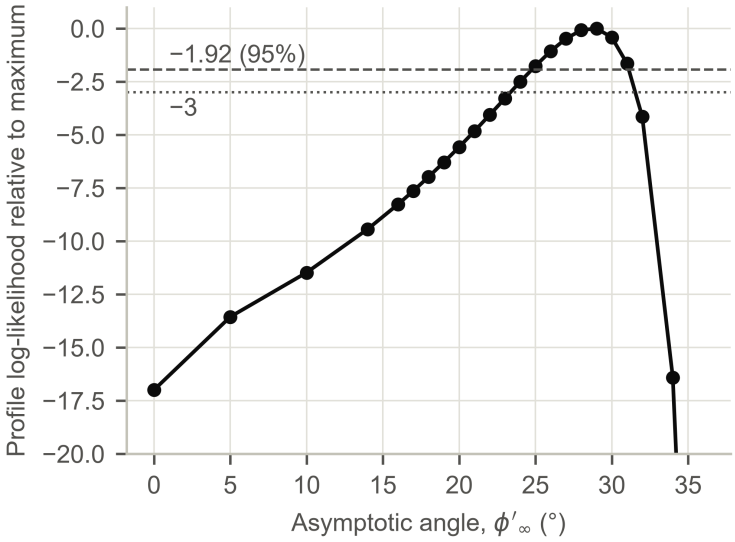


Fig. S3. Profile of the observed-data log-likelihood (censoring and truncation included) of the asymptotic angle ϕ'_∞ relative to its maximum (797.48); at each angle ζ_0, the fixed effects, and the variance parameters were refitted. ϕ'_∞ = 0 is the form without an asymptote (approximate: maximum over two values of ζ_0). Dashed: 1.92 units (approximate 95% interval); dotted: 3 units. First-order approximation; see Table S3.

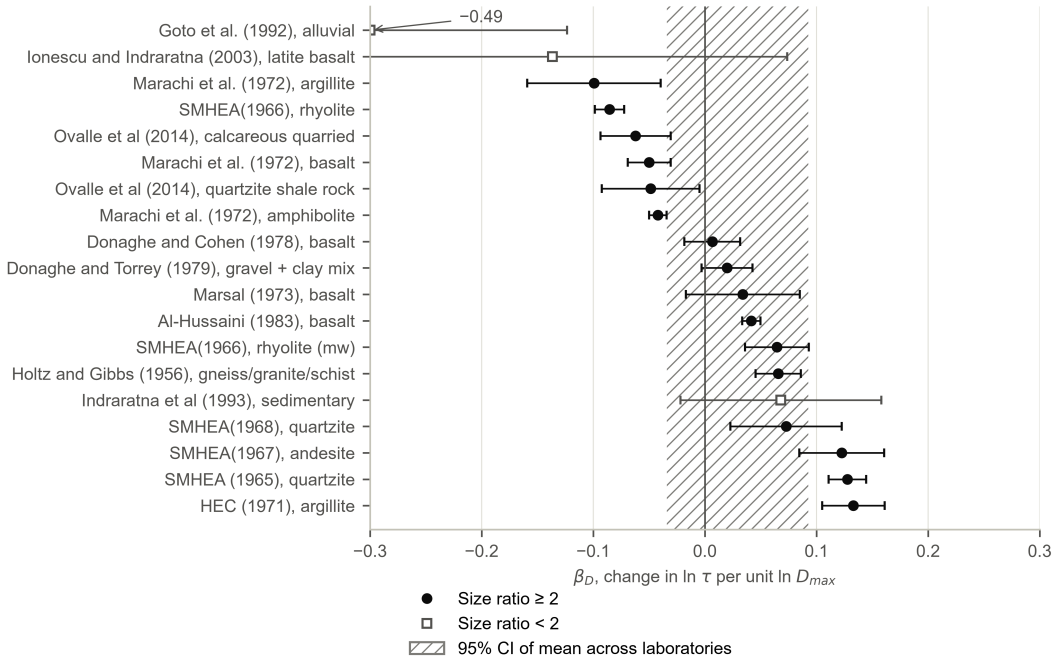


Fig. S4. Within-material specimen-size coefficient β_D by laboratory (bars: 95% confidence intervals; set intercepts adjusted for void ratio). Hatched band: Knapp–Hartung 95% confidence interval of the random-effects mean across 7 laboratories with size ratio ≥ 2 (+0.029; −0.034 to +0.093); between-laboratory standard deviation τ_D = 0.066.

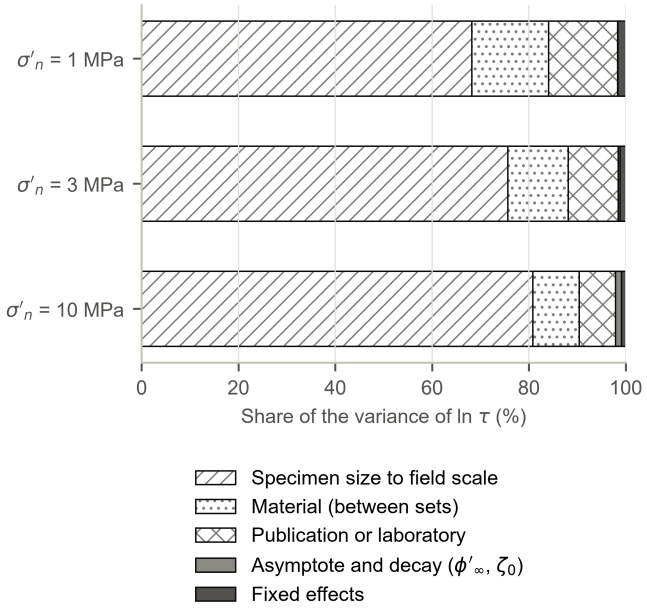


Fig. S5. Shares of the variance of the mean-envelope ln τ for an untested material of low rock quality placed at e = 0.65, scaled from 100 mm to 1 m maximum particle size, at three normal stresses.

TABLE S1. Mode B: every procedure evaluated (held-out tests at $\sigma'_n \geq 2$ MPa predicted from tests at $\sigma'_n \leq 1.5$ MPa of the same set)

Procedure	80% coverage	Below P05	Below P10	Mean z	CRPS	Δ CRPS	Holm p
Prior, no site tests	88%	4%	12%	−0.20	0.0455	+0.0097	1.00
Least-squares fit of ϕ'_1, m = 0.90	82%	6%	18%	−0.77	0.0433	+0.0075	1.00
Least-squares fit of ϕ'_1, m from training data	84%	2%	6%	+0.13	0.0305	−0.0053	1.00
Joint update of both random effects	76%	14%	24%	−0.64	0.0333	−0.0025	0.75
Level update of each set (Eq. 8)	80%	14%	20%	−0.56	0.0358	—	—
Level update, flagged sets discarded	80%	12%	20%	−0.54	0.0361	+0.0003	1.00
Flagged sets discarded, shared laboratory effect	74%	14%	26%	−0.62	0.0386	+0.0028	1.00

Note: 50 held-out tests at $\sigma'_n \geq 2$ MPa on 39 sets from 6 laboratories (11 publications), predicted from tests at $\sigma'_n \leq 1.5$ MPa of the same set. Mean z: mean standardized residual (negative: strength overpredicted). The procedures were developed while examining earlier versions of the model on related held-out sets, so the comparisons are exploratory. p-values from exact sign-flip tests on laboratory means, Holm-adjusted within Mode B. CRPS in units of ln τ; Δ CRPS relative to the level update. In the design procedure, the lower of the level-updated and prior percentiles is used above the largest σ'_n of the site tests.

TABLE S2. Secant friction angle ϕ'_s (°) of the waste rock of the example, median / 5th percentile of the mean envelope

Material	Laboratory, 1 MPa	Placed, 1 MPa	Placed, 2 MPa
Rock type A	41.9 / 37.7	37.2 / 30.5	35.5 / 29.4
Rock type B	43.5 / 39.1	38.9 / 32.0	36.8 / 30.4
Rock type C	40.6 / 36.4	37.0 / 29.7	35.4 / 28.5
Rock type D (untested)	41.5 / 37.3	37.1 / 30.1	35.5 / 29.0
Rock type E (untested)	41.5 / 37.3	35.1 / 28.3	33.8 / 27.5
Dump average, model correlation	42.0 / 39.3	37.4 / 31.3	35.7 / 29.8
Dump average, full correlation	42.0 / 37.7	37.4 / 30.5	35.7 / 29.2

Note: Laboratory: specimen void ratio and size. Placed: void ratio 0.65 (rock type E, 0.80), maximum particle size 1 m, size effect with zero mean. Each rock type is represented by a new set from the site laboratory. The full-correlation case sorts the samples of each rock type at every stress, an upper bound on dependence.

WORKED EXAMPLE OF THE UPDATE

The example follows Eqs. (3)–(9) of the paper at the maximum-likelihood point of the ridge ($\phi'_\infty = 29°$, $\zeta_0 = 0.428$), with the published parameters, and is checked against the implementation,

TABLE S3. Structural comparison of mean and covariance forms (data without censoring)

Model	Mean	Random-effect covariance	Parameters	Log-likelihood	Out-of-sample log score
M1	Bounded	Stress-decreasing (gradient of the mean)	17	956.7	946.8
M2	Bounded	Additive (level and slope)	17	953.5	943.9
M4b	Quadratic by class	Stress-decreasing (from M1)	17	953.0	940.5
M3b	Quadratic by class	Additive	17	949.6	937.4
M4	Quadratic	Stress-decreasing (from M1)	15	945.5	935.7
M3	Quadratic	Additive	15	940.9	931.2

Note: 741 tests without censoring or truncation; all models include rock quality class, void ratio, and ln Dmax. Log-likelihood: maximum likelihood with the exact Gaussian marginal for each publication, given the covariance structure; for M1 and M2, maximum over ϕ'_∞ from 20° to 28°. Out-of-sample: joint log score leaving out one publication at a time with the variance parameters fixed. First-order minus Laplace log-likelihood of M1: 1.1 at 20°, 1.8 at 22°, 3.0 at 24°, 3.9 at 25°, 3.8 at 26°, 9.9 at 28°; the Laplace maximum was 954.2. Largest single-publication contribution to M1 − M3: 7.2 of 15.8.

TABLE S4. Mode A with censored tests: fractions below the 5th and 10th percentiles

Band and form	Below P05	Below P10	Mean z
All stresses (741 tests, 55 censored, 23 laboratories)			
HiBER, bounded	4.9%–5.8%	8.4%–9.7%	+0.06
HiBER, quadratic	4.7%–5.4%	7.2%–8.2%	+0.09
HiBER, power	4.6%–5.3%	7.0%–8.1%	+0.11
Leps 2.0, hierarchical dispersion	3.4%–3.6%	4.5%–5.0%	+0.31
Leps 2.0, constant dispersion	4.0%–4.5%	5.7%–6.2%	+0.22
$\sigma'_n \geq 2$ MPa (146 tests, 28 censored, 8 laboratories)			
HiBER, bounded	6.8%–7.5%	11.0%–12.3%	−0.05
HiBER, quadratic	4.1%	5.5%	+0.17
HiBER, power	4.1%	5.5%	+0.24
Leps 2.0, hierarchical dispersion	4.8%	6.2%	−0.09
Leps 2.0, constant dispersion	4.1%	6.2%	−0.26
$\sigma'_n \geq 3$ MPa (90 tests, 24 censored, 5 laboratories)			
HiBER, bounded	6.7%	7.8%	+0.07
HiBER, quadratic	3.3%	3.3%	+0.24
HiBER, power	3.3%	3.3%	+0.33
Leps 2.0, hierarchical dispersion	3.3%	4.4%	−0.10
Leps 2.0, constant dispersion	3.3%	4.4%	−0.29

Note: Censored tests (axial strain at the maximum deviator ≥ 19%) are lower bounds of strength. Ranges: lower value counts a censored test as below a percentile only if its lower bound is; upper value counts it always. Mean z: mean standardized residual of the uncensored tests. Nominal: 5% and 10%.

which also averages over the ridge and the posterior of the variance parameters. Material: low rock quality, specimen void ratio 0.35, maximum particle size 178 mm; placement void ratio 0.65 and field size 1 m.

Step 1: Site tests. Three drained tests at σ'_3 = 0.3, 0.7, 1.2 MPa failed at $\sigma'1$ = 1.71, 3.42, 5.34 MPa. With the conversion of Cylwik et al. (2025) and m = 0.883 (the local slope of the prior at 1 MPa), they give σ'_n = 0.544, 1.228, 2.060 MPa and τ = 0.534, 1.076, 1.679 MPa, that is, x = -0.622, 0.192, 0.710 and y = -0.640, 0.060, 0.505.

Step 2: Conflict check. The prior mean at the tests is f = -0.673, 0.039, 0.500, the residuals are r = 0.032, 0.021, 0.005, and σ_ε = 0.041, 0.035, 0.031. With V from Eq. (7) built with the matrices below, D^2 = 0.15 with 3 degrees of freedom (p = 0.986): no conflict.

Step 3: Cut update. At the centroid $\bar{x}$ = 0.093, the gradient a stacks the gradient of the envelope with respect to (η, ζ) for the laboratory and the set effects and with respect to the fixed effects; $g_0 = \mathbf{a}^T\mathbf{Pa}$ = 0.0116. Eq. (8) gives $\hat{q}_0$ = 0.0167 with variance 0.00040. The shift $\mathbf{Pa}\hat{q}_0/g_0$ raises η by 0.0076 through the laboratory effect and by 0.0085 through the set effect, and moves the fixed effects slightly.

Step 4: Rock type, density, and size. The rock type is represented by a new set from the same laboratory: the updated laboratory and fixed effects are kept, and the set effect is drawn again from $\mathbf{G}_s$. The void ratio is then changed to the placement value through η, and the size shift adds a variance of $(\ln(D_f/D_l))^2(SE^2 + \tau_D^2)$ with zero mean. Above the largest σ'_n of the tests (2.06 MPa), the lower of the updated and prior percentiles is used.

Published matrices. Fixed effects β = (0.2036, 0.0872, 0.0280, −0.4879, 0.0216, −0.0675, −0.1005) in the order of Table 1 (β_c good, moderate, low; β_e; κ; δ_c moderate, low). Dispersion: ln σ_ε = -3.316 −0.207x.

TABLE S5. Worked example: secant friction angle ϕ'_s (°), median / 5th percentile of the mean envelope

Stage	0.5 MPa	1 MPa	3 MPa	5 MPa
Prior, laboratory	43.9 / 38.2	41.3 / 36.3	38.0 / 33.5	36.7 / 32.4
Updated (rock type), laboratory	44.1 / 39.1	41.6 / 37.2	38.2 / 34.1	36.9 / 32.9
Prior, placed and field size	39.1 / 31.3	37.2 / 30.1	34.9 / 28.4	34.0 / 27.8
Updated (rock type), placed and field size	39.3 / 32.0	37.4 / 30.6	35.1 / 28.8	34.2 / 28.1
Design 5th percentile	32.0	30.6	28.4	27.8
Implementation (envolvente_diseno)	39.3 / 32.0	37.4 / 30.6	34.8 / 28.4	33.9 / 27.7

Note: Plug-in values at the maximum-likelihood point; the implementation averages over the ridge of (ϕ'_∞, ζ_0) and the posterior of the variance parameters (conflict p = 0.984).

Covariance of the fixed effects, $\mathbf{C}_\beta \times 10^4$

	β good	β moderate	β low	β_e	κ	δ moderate	δ low
β good	12.07	7.58	5.23	−12.28	0.39	0.04	0.17
β moderate	7.58	8.71	4.65	−10.58	0.27	−0.77	0.09
β low	5.23	4.65	6.42	−7.51	0.46	0.05	−1.06
β_e	−12.28	−10.58	−7.51	23.23	0.39	0.01	−0.12
κ	0.39	0.27	0.46	0.39	0.75	0.04	0.00
δ moderate	0.04	−0.77	0.05	0.01	0.04	5.00	0.75
δ low	0.17	0.09	−1.06	−0.12	0.00	0.75	5.92

$\mathbf{G}_p$ and $\mathbf{G}_s$ (rows and columns η, ζ), $\times 10^4$

	η (laboratory)	ζ (laboratory)	η (set)	ζ (set)
η	53.65	−4.68	59.56	−20.21
ζ	−4.68	61.64	−20.21	78.26

TABLE S6. Factor of safety of dumps on a competent level foundation (Morgenstern–Price, Slide2), placed waste rock: median / 5th-percentile envelope

Envelope	150 m	400 m	700 m
Dump average, model correlation	1.80 / 1.40	1.62 / 1.28	1.54 / 1.22
Dump average, full correlation	1.79 / 1.34	1.61 / 1.24	1.53 / 1.19
Weakest tested rock type only	1.76 / 1.29	1.59 / 1.20	1.52 / 1.15
Untested, degraded rock type E only	1.62 / 1.20	1.49 / 1.14	1.43 / 1.10

Note: Uniform dry face without benches; unit weight 17.2 kN/m^3 (rock type E only: 15.7 kN/m^3). Monte Carlo at 700 m with 600 envelope samples, each with its own search: model correlation 0 failures (lognormal reliability index 3.2); rock type E only 5 (index 2.3). 700 m, model correlation: without size shift 1.54 / 1.40; laboratory density with size shift (21 kN/m^3) 1.74 / 1.38; laboratory density and size 1.74 / 1.59; size effect at its estimated mean 1.62 / 1.29; Weibull scaling 1.40 / 1.12; independent size shifts per rock type 1.55 / 1.35; tested samples taken as representative 1.51 / 1.22; quadratic form 1.49 / 1.19.